

3.1 – 3.5 Review Worksheet (In-Class)

1. State the domain for the function $f(x, y) = \frac{\sqrt{x - y^2}}{e^x - 2}$ and create a picture of the domain in the xy -plane.

2. Determine each of the following limits or show that it does not exist.

a) $\lim_{(x,y) \rightarrow (2,1)} \frac{xy}{x^2 + 3y^2}$

b) $\lim_{(x,y) \rightarrow (0,0)} \frac{xy}{x^2 + 3y^2}$

c) $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 \sin^2(y^3)}{x^4 + y^{12}}$

3. Suppose that a function $f(x, y)$ is continuous over all of $\mathbb{R}^2$ and has the following properties. First suppose that for any point (a, b) it is the case that $f(a, b) = -f(b, a)$. Second suppose that the following points exist on the surface created by $f(x, y)$:

$$(1, 2, 3), (2, 5, 10), (3, -4, 5), (-1, 5, 2), (4, 7, 1), (-4, -3, 1)$$

Use this information to determine the exact value of the limit $\lim_{(x,y) \rightarrow (-4,3)} f(x, y)$.

4. Compute each of the indicated partial derivatives for the function $f(x, y) = x^2 y^3 + \frac{y+x}{\sin(x)}$.

a) Find f_x .

b) Find f_{yx} .

5. Like in the last question, consider the function $f(x, y) = x^2 y^3 + \frac{y+x}{\sin(x)}$.

a) Show that the function is differentiable at the point $\left(\frac{\pi}{2}, \pi\right)$.

b) Find a point where we cannot guarantee that the function is differentiable.

6. Determine $\frac{\partial z}{\partial y}$ for the surface given by the equation $xz + yz^2 = \ln(y + z + x)$

7. Suppose that $z = x^4 + x^2y$, $x = 2t + 5$, $y = \sin(t) - e^{2t}$. Find $\frac{dz}{dt}$ when $t = 0$.

8. Suppose that $z = \frac{x}{2y+x}$, $y = pq\sqrt{r}$, and $x = p\sqrt{qr}$. Find $\frac{\partial z}{\partial p}$ when $p=2$, $q=1$, $r=4$.
9. A manufacturer has modeled its yearly production function P (the value of its entire production in millions of dollars) by the function $P(L, K) = 1.47L^{0.65}K^{0.35}$. L represents the number of labor hours (in thousands) and K represents the invested capital (in millions of dollars). Suppose that when $L = 30$ and $K = 8$, the labor force is decreasing at a rate of 2000 labor hours per year and the capital is increasing at a rate of \$500,000 per year. Find the rate of change of production. Round your final answer to 3 decimal places.